

■ AIFAA Backend Development Roadmap (8 Weeks)

Tech Stack

Language: Python 3.x

Framework: FastAPI

Database: PostgreSQL (SQLAlchemy + Alembic)

Auth: JWT (PyJWT or fastapi-users)

AI Layer: OpenAI API or local inference endpoint

Tools: Pydantic, HTTPX, dotenv, logging

Week 1 – Setup & Environment

- Initialize Git repo and virtual environment
- Install dependencies (FastAPI, SQLAlchemy, Alembic, etc.)
- Setup `.env`, config, and FastAPI base structure
- Initialize Alembic and connect to PostgreSQL
- Commit first working version

Week 2 – Database Models & Migrations

- Define User, Language, FirstAidGuide, EmergencyCallLog, ChatSession, Message, Feedback, LocationLog models
- Configure relationships and Alembic migrations
- Run ``alembic upgrade head``
- Verify database schema and seed initial data

Week 3 – Authentication & User Management

- Implement JWT auth or FastAPI Users
- Add ``/auth/register``, ``/auth/login``, ``/auth/me`` routes
- Secure endpoints with ``Depends(get_current_user)``
- Write integration tests for authentication

Week 4 – Core APIs (CRUD + Data Layer)

- Add routes for guides, languages, feedback, emergency logs
- Implement Pydantic schemas for input/output
- Add pagination and exception handling

Week 5 – AI & Chat Layer

- Integrate OpenAI or local inference endpoint
- Create ``/chat`` routes with chat sessions and messages
- Implement message persistence and async calls

Week 6 – Location & Emergency Features

- Add `/location/update` and `/emergency/` endpoints
- Log user location and emergency events
- Integrate reverse-geocoding or responder logic

Week 7 – Testing, CI/CD & Optimization

- Write unit and integration tests with pytest
- Add GitHub Actions pipeline for linting, testing, and migrations
- Optimize logging and error handling

Week 8 – Deployment & Documentation

- Prepare production `.env` and Dockerfile
- Deploy FastAPI backend with Supabase/PostgreSQL
- Set up Alembic migrations on deploy
- Write README and API docs